

Day 3 – Shell Scripting: File Checks & Conditions

Objective:

Learn file and directory checks, permission validation, and defensive scripting for production use.

1. File Test Operators:

- f : Checks if file exists
- d : Checks if directory exists
- e : Checks if file or directory exists
- s : Checks if file exists and is not empty

2. Permission Checks:

- r : Read permission
- w : Write permission
- x : Execute permission

3. Logical Operators:

- && : AND condition
- || : OR condition
- ! : NOT condition

4. Practice Scripts Covered:

- File existence and readability check
- Directory existence and listing contents
- Nginx access log validation and tail last 5 lines

Example Production Script (Nginx Log Check):

```
LOG_FILE="/var/log/nginx/access.log"
if [ ! -f "$LOG_FILE" ]; then
    echo "Nginx not installed or access log missing"
    exit 1
fi
tail -n 5 "$LOG_FILE"
```

Best Practices:

- Always validate arguments
- Always quote variables
- Use exit codes on failure
- Never assume file existence in production scripts

Day 3 completed successfully